Development of a Disaster-Resilient Infrastructure Preparedness Chatbot for Flood-Prone Communities

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***Abstract*—Accessing clear, accurate, and actionable guidance on infrastructure preparedness remains a major challenge for flood-prone communities in the Philippines, limiting local capacity to mitigate structural damage prior to heavy rainfall. In alignment with Sustainable Development Goal 9 (Industry, Innovation, and Infrastructure), this paper proposes the development of a localized chatbot that leverages Retrieval-Augmented Generation (RAG) to deliver grounded guidance on pre-flood infrastructure protection, drainage maintenance, and risk minimization. The system integrates a curated knowledge base of authoritative government guidelines and advisories from the Department of Science and Technology – Philippine Atmospheric, Geophysical and Astronomical Services Administration (DOST-PAGASA) and the Office of Civil Defense / National Disaster Risk Reduction and Management Council (OCD/NDRRMC). Utilizing Meta's Llama 3.1 8B Instant model deployed on the Groq LPU API platform, the chatbot synthesizes technical guidelines into clear recommendations across English, Tagalog, and Taglish queries. System evaluation is structured around a benchmark test set measuring response correctness, relevance, groundedness, and boundary recognition for out-of-scope inquiries. This project provides a practical model for AI-assisted information access in community-level disaster risk reduction.**

Keywords—Retrieval-Augmented Generation, Flood Preparedness, Infrastructure Resilience, Disaster Risk Reduction, Llama 3.1, SDG 9

I. Selected UN SDG and Justification

The selected Sustainable Development Goal for our discussion is SDG 9 – Industry, Innovation, and Infrastructure. The suggested project is called "Development of a Disaster Resilient Infrastructure Preparedness Chatbot for Flood Prone Communities." The idea of the project is to create an accessible informational source that will give communities some understanding of how they can prepare their infrastructure for flooding[1].

SDG Target 9.1 promotes development of high-quality, reliable, sustainable, and resilient infrastructure. Specifically, our project will be focused on developing disaster resilient infrastructure and preparation of communities for floods. Flooding can impact infrastructure in many ways. Roads, buildings, drainage systems, bridges, and many other infrastructure elements that communities rely on can be affected by floods. Providing communities with relevant information about infrastructure preparedness can be helpful in preventing any damages[1].

Yet, flooding can lead to significant damage to infrastructure especially in those communities that are prone to flooding. Floods can impact housing, transportation, infrastructure and other community assets. Indicate that knowledge of flash floods, their perceived severity and vulnerability, as well as past experience with floods, are among the factors related to the intention of Filipinos to prepare for flash floods. This research also emphasizes the need to have access to information on how to mitigate flooding and how to prepare for them. Furthermore, some communities might encounter challenges when obtaining accurate and comprehensible information on how to prevent damage to infrastructure, identify possible risks, maintain it and prepare it for the case of floods. Thus, SDG 9 is also applicable to the proposed project since the development of resilient infrastructure is not limited by physical aspects alone, but also by information and preparation. In this respect, a chatbot will be useful to solve the problem of information access[2].

The proposed system will apply the method called Retrieval-Augmented Generation (RAG) which is a system that incorporates the processes of information retrieval and language generation, whereby the relevant information is retrieved from an external knowledge source before the process of generating a response. RAG is a solution that involves the combination of a language model and an external knowledge source, thereby making it possible for the model to use the retrieved information in answering knowledge-intense questions. This solution is suitable for the proposed chatbot as it will make it possible for the system to retrieve information from a curated database of reputable sources including government guidelines, disaster preparedness documents, infrastructure suggestions, among others. Questions related to flood prevention, infrastructure preparation, maintenance activities, and risk assessment can be answered using the generated responses. RAG is especially important for domain-specific tasks since it makes it possible for a system to use external information rather than just the information contained in the language model[3].

II. Narrow Problem Statement

Level	Group Output
Broad domain	Infrastructure resilience and disaster preparedness

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SDG-related issue	SDG 9 – Industry, Innovation and Infrastructure, particularly Goal 9's focus on building resilient infrastructure. SDG 9 promotes the development of reliable, sustainable, and resilient infrastructure that supports human well-being and development.
Target users	Residents and community members living in communities affected by floods in the Philippines, especially those who require relevant and comprehensible information regarding how to minimize the risks associated with flooding.
Narrow problem	Lack of accessible and easy-to-understand information on how residents and communities can prepare for and protect basic infrastructure from flood-related damage. This includes information about drainage maintenance, infrastructure preparedness, flood-risk reduction, and actions that can be taken before and during periods of heavy rainfall.
Proposed chatbot focus	A chatbot that uses RAG for flooding infrastructures preparedness, using information from credible government, disasters preparedness, weather, and infrastructure sources. The bot will aid the user in answering their queries on flood preparedness, infrastructures protection, drainage, maintenance, and risk minimization.

Residents and community members who live in areas where there is a likelihood of flooding face challenges in accessing clear, accurate, and comprehensible information on how to prepare infrastructure prior to the occurrence of floods. Information on drainage systems maintenance, home and community facility protection, mitigation of flood risks, and infrastructure preparation can be found in various government and disaster-preparation documentation and it can be challenging for people to access the necessary information quickly. This project will address this information problem by developing a Retrieval-Augmented Generation (RAG) chatbot that retrieves information from a curated collection of reliable documents and provides simple answers to questions about pre-flood infrastructure preparedness. The problem is worth investigating because improving access to practical preparedness information may help communities better understand recommended actions for reducing infrastructure-related flood risks.

III. Primary Research Question

"What is the degree of success for a chatbot that uses RAG to retrieve accurate answers on topics related to flood-preparedness and infrastructure protection from a curated set of documents?"

This research question directly addresses the identified information problem: the difficulty of accessing clear and reliable information about preparing for and protecting infrastructure from flooding. It is focused because the chatbot will be evaluated only on flood preparedness and infrastructure-protection questions rather than general disaster-related topics. It is also measurable because the group can evaluate the chatbot using criteria such as answer correctness, relevance, and groundedness in the retrieved documents. RAG is appropriate for this problem because it allows a language model to retrieve relevant information from a defined knowledge base before generating an answer, helping ground responses in the provided sources rather than relying only on the model's internal knowledge. Research on RAG evaluation also emphasizes the importance of assessing both the retrieved context and the generated answer for reliable question answering.

IV. Initial Knowledge-Base Data Sources

Information	DOST-PAGASA	OCD/NDRRMC
Source name or organization	Philippine Atmospheric, Geophysical and Astronomical Services Administration (DOST-PAGASA)	Office of Civil Defense (OCD) / National Disaster Risk Reduction and Management Council (NDRRMC)
Document type	Government webpages, flood forecasts, flood bulletins, advisories, and learning materials	Government plans, disaster preparedness guidelines, reports, advisories, and risk-reduction publications
Relevance	PAGASA provides official information about flood forecasting,	OCD/NDRRMC provides information on disaster preparedness,

	flood advisories, rainfall conditions, and flood-warning guidance. These materials can help the chatbot answer questions about flood risks, warning levels, and precautionary actions. PAGASA's flood information can also help users understand the appropriate actions to take when flooding is expected.	risk reduction, emergency planning, and community-level actions. The NDRRMP 2020–2030 provides a national framework for disaster risk reduction and management in the Philippines. These materials can support questions about what communities should do before, during, and after flooding.
Availability	Highly accessible. PAGASA provides flood information, forecasts, advisories, and learning resources publicly through its official website. The group can select and download relevant materials for the RAG knowledge base.	Accessible. OCD/NDRRMC publishes disaster-related plans, guidelines, and resources through its official platforms. The NDRRMP 2020–2030 and other relevant disaster preparedness materials are publicly available.
Reliability and permission	Highly authoritative**.** PAGASA is a Philippine government agency under the Department of Science and Technology (DOST) and is responsible for weather, hydrological, and flood forecasting and warning services. The group will retain the original source information and properly cite the documents used in the knowledge base.	Highly authoritative. OCD/NDRRMC are official Philippine government disaster-risk-management bodies. The NDRRMP is an official national planning document, making it an appropriate source for disaster preparedness and risk-reduction information. The group will properly identify and cite the documents included in the knowledge base.

V. Initial Evaluation Data Plan

To evaluate the degree of success of the proposed Retrieval-Augmented Generation (RAG) chatbot, a preliminary benchmark test question set will be constructed to measure answer correctness, relevance, and groundedness against retrieved context [3], [4].

Realistic user questions will be gathered primarily from inquiries raised during local Barangay Disaster Risk Reduction and Management (BDRRM) community orientations, public flood preparedness FAQs, and official disaster advisories published by DOST-PAGASA and OCD/NDRRMC. The evaluation dataset will encompass diverse question types to thoroughly test the system. Factual and informational queries will address specific guidelines, such as standard heights for elevating electrical installations. Procedural queries will cover actionable maintenance steps, including neighborhood drainage cleaning prior to heavy rainfall. Complex scenario-based queries will incorporate factors like community vulnerability and perceived flood severity [2] to evaluate retrofitting advice for specific housing structures. Furthermore, localized conversational phrasing in Tagalog and Taglish will be included to evaluate the chatbot's performance with authentic community language inputs [2].

Ground truth reference answers and supporting passages will be established by pairing each test questions with specific text chunks extracted directly from official DOST-PAGASA advisories and OCD/NDRRMC planning documents. Ideal benchmark responses will be manually formulated strictly from these verified passages to evaluate retrieval accuracy and generated answer precision [3], [4]. To evaluate boundary recognition and prevent harmful hallucinations, the test set will explicitly include out-of-scope questions, such as requests for live weather forecasts or individual engineering certifications [4]. The evaluation will verify whether the system correctly identifies missing context and issues and appropriate fallback response directing users to official agencies [4]. Actual data collection and manual annotation for this evaluation design will be conducted in subsequent project phases.

VI. Possible Language Model and Justification

The proposed language model for the generation component of the RAG pipeline is Meta's Llama 3.1 8B Instant (llama-3.1-8b-instant), deployed via the Groq LPU API platform [5]. Factual adherence is critical when delivering disaster preparedness and infrastructure mitigation advice to flood-prone communities [2]. Llama 3.1 8B Instant is specifically optimized for instruction-following taks and precise context alignment, making it well-suited for adhering to strict system prompts that constrain generated outputs strictly to retrieved context

passages [5]. It's 8-billion parameter size and expanded 128k context window provide a favorable balance between deep language comprehension, multi-document synthesis, and rapid response generation for public end-users [5].

In the proposed RAG architecture, the model will receive user queries alongside relevant context passages retrieved front the DOST-PAGASA and OCD/NDRRMC knowledge base [3]. Its expected role is to synthesize complex technical guidelines into clear, non-technical, and actionable guidance tailored for community residents without introducing unverified external claims [3], [5].

From a practical perspective, using Llama 3.1 8B Instant via the Groq API provides ultra-low latency inference while eliminating local GPU hardware constraints and high operational costs for academic research [5]. The open-weights foundation allows for zero-shot prompt engineering or fine-tuning, while its multilingual capabilities facilitate effective processing of Tagalog and Taglish inputs [2], [5]. This selected model represents a preliminary proposal for project prototyping, its performance will be empirically compared against alternative open-source language models during system implementation.

VII. Group Reflection

The proposed problem is narrow and manageable enough because the chatbot will focus specifically on flood preparedness and disaster-resilient infrastructure information for flood-prone communities rather than covering all types of disasters or infrastructure concerns. The group can create a focused knowledge base using a manageable collection of reliable government guidelines, disaster-preparedness documents, and infrastructure-related materials. The chatbot will also be evaluated using a limited set of test questions about topics such as infrastructure preparation, maintenance, flood-risk reduction, and preventive measures. By limiting the scope to pre-flood preparedness and using RAG to provide answers based on the selected documents, the project can be developed, tested, and evaluated within the available time and resources of a student research project.

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